

Agroforestry Biomass Supply Chain Design

A MILP Model with Age-Lag Constraints for Product Cascading

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Overview

- ① **Introduction** — Motivation & Research Gap
- ② **Literature Review** — AFS Systems, Biomass Growth, Supply Chains
- ③ **MILP Model** — Sets, Variables, Objective, Constraints
- ④ **Case Study** — Central European Poplar-based AFS (Saxony-Anhalt)
- ⑤ **Optimization Results** — Supply Chain Configuration & Flows
- ⑥ **Conclusion** — Contributions & Open Questions

Section 1

Introduction

Motivation

- Temperate **agroforestry systems (AFS)** integrate fast-growing trees (poplar, willow) into arable land via alley-cropping
- Woody biomass from AFS is a **sustainable feedstock** for bio-based value chains — preferred over pure energy use as biogenic carbon is stored in products
- In Germany: ~1,200 ha of silvoarable AFS in 2025 — only **0.02 %** of agricultural land despite substantial ecological benefits
- **Core challenge:** coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains

Research question

How can integrated MILP models link **age-dependent stand dynamics** at field sites to **regional supply chain design** for multi-product AFS biomass?

Research Gap & Contributions

Prior literature on biomass supply chains (MILP) typically:

- treats biomass availability as **exogenous input**
- focuses on a **single downstream value chain** (e.g. bioenergy)
- ignores **age-lag constraints** linking consecutive harvests

This paper contributes:

- ① Novel **MILP formulation** (AFS-SCD) with explicit age-lag constraints enforcing minimum/maximum rotation ages
- ② **Allometric yield functions** (Gompertz) for age-specific, component-specific biomass fractions (stems, branches, residues)
- ③ **Multi-product, multi-period network design** with product quality cascades and storage
- ④ Application to a **real-world case study** in Central Germany

Section 2

Literature Review

Agroforestry System Types

Short-rotation coppice (SRC)

- Monoculture, 8,000–20,000 cuttings/ha
- Rotation: 3–7 years, stand lifetime: 20–25 years
- Root system regenerates after harvest — no replanting
- Benchmark for biomass production

Short-rotation AFS (SRAFS)

- SRC-type tree strips in alley-cropping arrangement
- Trees cover 10–30 % of parcel area
- Boundary effect: higher light & water higher individual-tree biomass

Long-rotation AFS (LRAFS)

- Same spatial structure as SRAFS
- Rotation cycles: 12–15+ years
- Larger stem diameter high-value timber (veneer, sawlogs, engineered wood)
- Timber-quality species: walnut, cherry, oak, high-grade poplar

Key insight

SRAFS and LRAFS are not sharply separated — **intended use, establishment decisions, and harvest timing** jointly determine economic viability.

Biomass Growth Models

Gompertz growth function for above-ground biomass $B_{AG}(t)$:

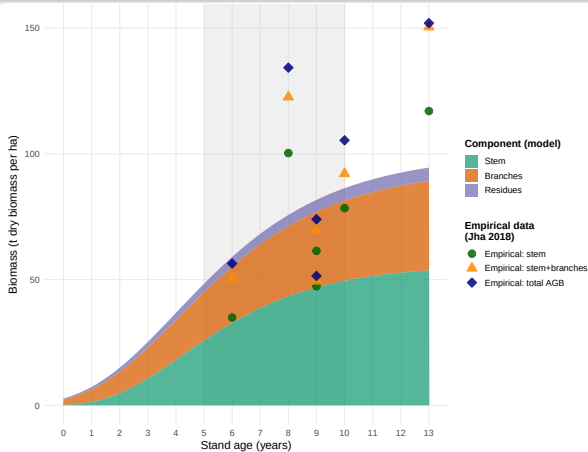
$$B_{AG}(t) = A \exp(\exp(k(t - t_0)))$$

Component-specific fractions (stems $p = 1$, branches $p = 2$, residues $p = 3$):

$$B_{AG,p}(t) = B_{AG}(t) \frac{h_p(t)}{h_1(t) + h_2(t) + h_3(t)}$$

- Parameters calibrated from SRAFS field trials in Southern Germany (Huber et al., 2018)
- Age-specific yield coefficients p,a pre-computed and embedded in the MILP (constraints C3)

Gompertz Growth Curves — Poplar Max 3



Key observations:

- Stem share rises from ~55 % at age 4 to >75 % at age 10
- Growth peak (inflection) at ~age 4
- Total AGB at age 10: ~65 t dry matter/ha
- Component fractions $f_p(t)$ determine yield coefficients p_a in constraint C3

Figure 1: Aboveground biomass components (stems, branches, residues) by stand age for poplar clone Max 3 under SRAES conditions. Parameterisation:

Biomass Supply Chain Literature

Table 1: Literature gap summary

Aspect	State of the art	This paper
Biomass availability	Exogenous or simplified	Endogenous via Gompertz + age-lag
Rotation decisions	Not modelled	Binary z_{ist} with min/max age
Product types	Single or aggregated	Multi-product cascade $(p,p) \quad Q$
Network structure	Static	Multi-period, storage + processing
Case study scope	Generic	Real SRC sites, Saxony-Anhalt

Section 3

MILP Model (AFS-SCD)

Sets, Indices & Parameters

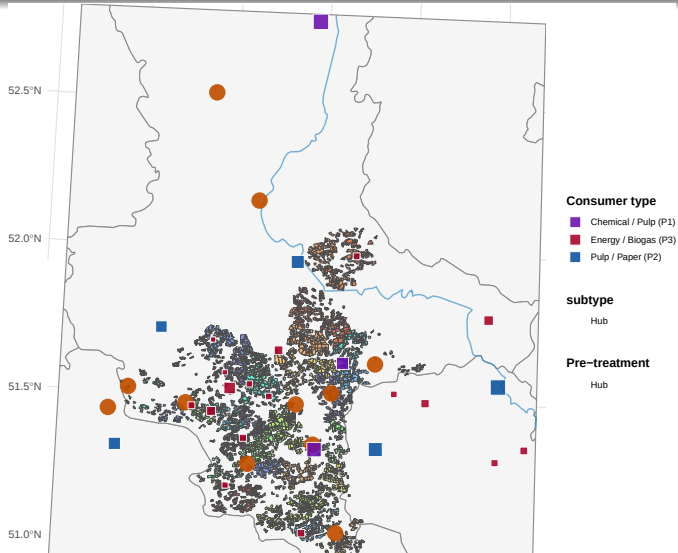
Sets & indices

Symbol	Meaning
I	AFS sites (field parcels)
J	Storage/processing facilities
K	Industrial consumers
P	Biomass product types
Q	Quality cascade pairs (p, p)
T	Planning periods
S	Feasible activity arcs (s, t)

Key parameters

Symbol	Meaning
p, a	Yield (t/ha) for product p at age a
$AREA_i$	Site area (ha)
c_i^{est}	Establishment cost (/ha)
c_i^{harv}	Harvest cost (/ha)
c_i^{opp}	Opportunity cost (/ha/yr)
R_{kp}	Revenue (/t) at consumer k
d_{ij}, d_{jk}	Transport distances (km)

Supply Chain Network Structure



Age-Lag Arc Structure $\mathcal{S}$

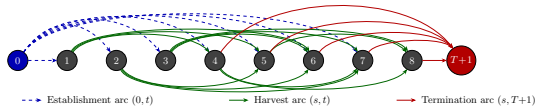


Figure 3: Arc network $\mathcal{S}$: establishment arcs $\mathcal{S}^{est}$, harvest arcs $\mathcal{S}^{harv}$ (feasible only for $A_{min} \leq t \leq A_{max}$), and termination arcs $\mathcal{S}^{term}$.

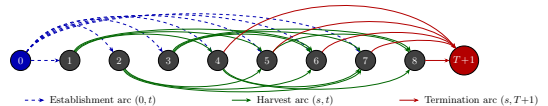


Figure 4: Arc network $\mathcal{S}$ (alternative view): consecutive harvest arcs enforce path-consistent rotation sequences across the planning horizon T .

Age-Lag Path Examples

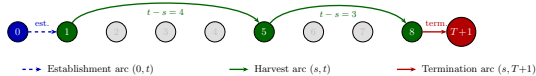


Figure 5: Example path 1: establishment at $t = 0$, first harvest at $t = 5$ (age 5), second harvest at $t = 9$ (age 4), termination at $t = 12$. All arcs satisfy $A_{min} = 3$, $A_{max} = 15$.

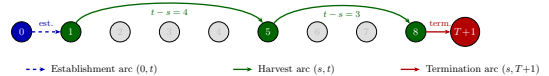


Figure 6: Example path 2: alternative rotation pattern with earlier first harvest. Path-connectivity constraint C2 ensures no orphaned harvest decisions.

Decision Variables

Variable	Type	Meaning
$z_{ist} \in \{0, 1\}$	Binary	Activity arc: site i from age s to t (establishment, harvest, or termination)
$X_{ijpt} \geq 0$	Continuous	Biomass flow (t): site i storage j , product p , period t
$X_{jkppt} \geq 0$	Continuous	Processed flow (t): storage j consumer k , cascade (p, p)
$S_{jpt} \geq 0$	Continuous	Inventory (t) at storage j , product p , end of period t

Objective Function

Maximise **total contribution margin** over the planning horizon T :

$$\max Z = \underbrace{R_{kp}}_{\substack{j,(p,p),t \\ \text{Revenue}}} \underbrace{X_{jkppt}}_{\substack{C^{est} \quad C^{opp} \quad C^{main} \\ \text{Site costs}}} \underbrace{C^{harv}}_{\substack{C^{tr-raw} + C^{tr-pre} \\ \text{Transport costs}}} + \underbrace{C^{stor}}_{\text{Storage costs}}$$

- **Opportunity costs** penalise foregone crop revenues for each year a site is under AFS
- **Two-stage transport** distinguishes raw biomass (site → storage) from pre-treated biomass (storage → consumer)
- **Storage costs** apply from year 2 onward (year-of-harvest drying is cost-free)

Constraints (C1–C4)

(C1) Single establishment per site:

$$\sum_{(0,t) \in S^{est}} z_{i0t} = 1 \quad i \in I$$

(C2) Path connectivity — flow conservation:

$$\sum_{(s,t) \in S} z_{ist} = \sum_{(t,u) \in S} z_{itu} \quad i, t$$

(C3) Biomass yield via Gompertz coefficients:

$$X_{ijpt} = \sum_{(s,t) \in S^{harv}} p(ts) \cdot AREA_i \cdot z_{ist} \quad i, p, t$$

(C4)/(C5) Transport linkage & inventory balance — follows OPTIMASS network-flow structure (De Meyer et al., 2014)

Constraints (C6–C8) & Cascade Logic

(C6) Storage capacity:

$$S_{jpt} \leq CAP_j^{stor} \quad j, t$$

(C7) Processing capacity:

$$X_{ijpt} \leq CAP_j^{proc} \quad j, t$$

(C8) Demand with quality cascade:

$$X_{j,(p,p)Q} \leq D_{kpt}^{max} \quad k, p, t$$

i Product cascade principle

$(p, p) \ Q$ allows a **higher-quality product** p (e.g. stems) to satisfy demand for a **lower-quality product** p (e.g. chips for energy) — operationalising cascading use as the central economic rationale

Section 4

Case Study

Study Region & Supply Chain Configuration

- **Region:** tri-state area of Saxony-Anhalt, northern Thuringia, northern Saxony (Germany)
- Highest density of registered SRC *Feldblöcke* in Germany
- **Supply chain structure:** AFS sites (sources) storage/processing nodes industrial consumers (sinks)

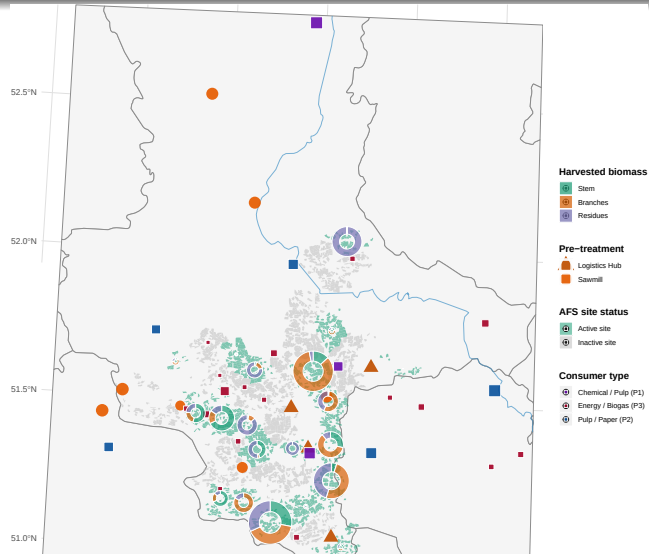
Sites *I*

- Registered SRC parcels from INVEKOS land-use register
- Clustered into representative site groups
- Site-specific area, establishment/harvest costs

Consumers *K*

- 3 biomass product types: stems (P1), branches (P2), residues (P3)
- Industrial consumers: paper/pulp, chemical biorefinery, energy plants
- Minimum demand threshold: 1.5 units/period

Supply Chain Network — Saxony-Anhalt



Consumer Demand & Product Classification

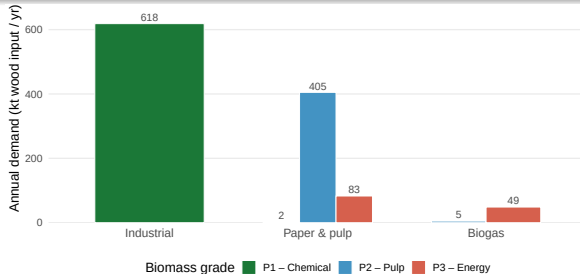


Figure 8: Consumer demand profile by product type (P1 stems, P2 branches, P3 residues) across industrial consumers $k \in K$. Demand values scaled to 1/5 of raw register values; consumers with total demand < 1.5 units excluded.

Product	Type	Primary use	Cascade target
P1 — Stems	High quality	Sawlogs, pulp, engineered wood	Can supply P2, P3 demand
P2 — Branches	Medium quality	Chips, biomass boards	Can supply P3 demand
P3 — Residues/twigs	Low quality	Energy (combustion,	End of cascade

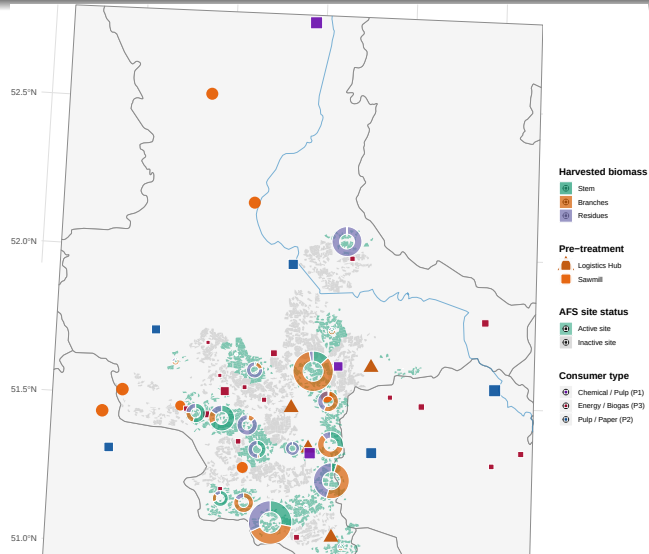
MILP Model Parameters

Parameter	Value	Source
Planning horizon $ T $	40 periods	—
Max. rotation age A_{max}	15 years	LRAFS benchmark
Min. rotation age A_{min}	3 years	SRAFS minimum
Raw biomass transport c^{tr-raw}	0.08 /(tkm)	Literature
Pre-treated transport c^{tr-pre}	0.06 /(tkm)	Literature
Establishment cost c^{est}	2,500 /ha	(Testa et al., 2014)
Harvest cost c^{harv}	150 /ha	Literature
Opportunity cost c^{opp}	45 /ha/yr	Literature
Solver	Gurobi 10, MIP gap 1 %	—

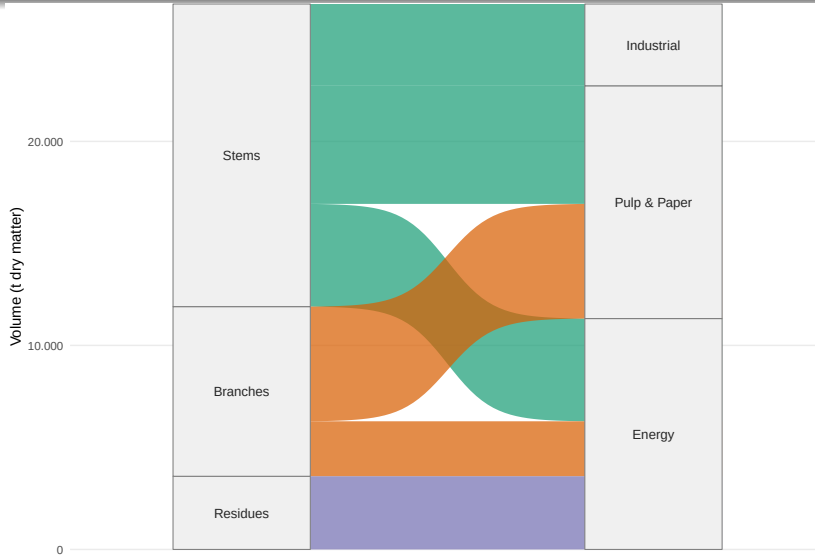
Section 5

Optimization Results

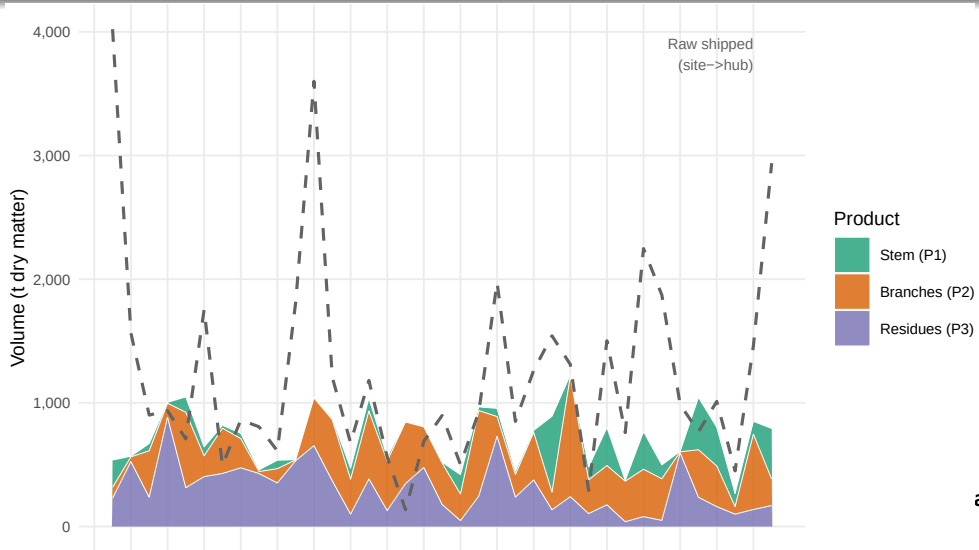
Supply Chain Network — Optimal Configuration



Harvest & Rotation Patterns



Product Mix & Biomass Flows



Inventory Dynamics at Storage Nodes

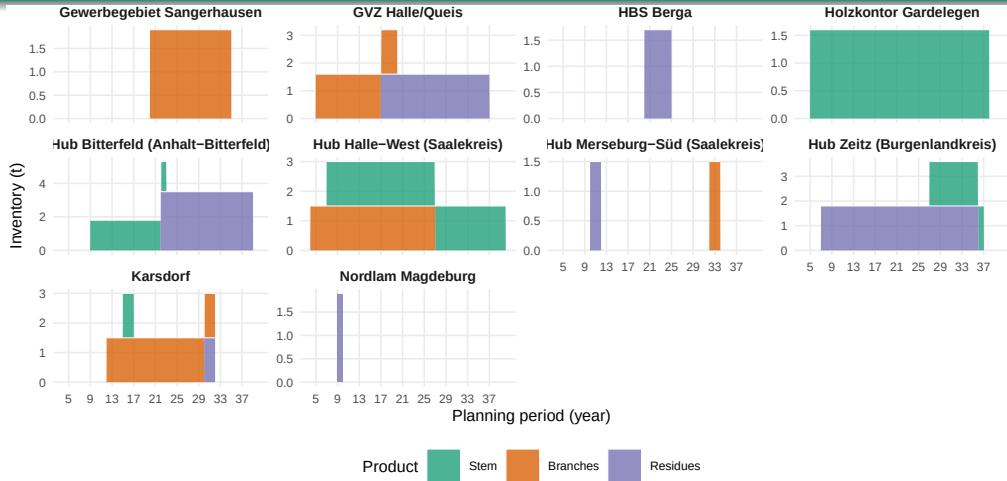


Figure 12: Inventory levels S_{jpt} at storage nodes by product type and period. Storage buffers harvest peaks (spring/summer) to ensure continuous supply to consumers. Year-of-harvest

Section 6

Conclusion

Contributions & Key Findings

Novel MILP formulation (AFS-SCD): explicit age-lag constraints link establishment and consecutive harvest decisions — first integrated AFS supply chain model of this type

Allometric Gompertz yield functions embedded directly in the MILP via pre-computed p, a coefficients — no exogenous biomass input needed

Multi-product cascade operationalises cleaner production principle: high-quality material use before energy recovery

Real-world case study (Saxony-Anhalt) demonstrates feasibility at regional scale

Open Research Questions

! Planned computational experiments

- **Allometric parameter uncertainty:** sensitivity of optimal rotation length to Gompertz parameter ranges
- **Price scenarios:** impact of biomass price trajectories (P1/P2/P3) on product mix and supply chain configuration
- **Policy incentives:** effect of AFS subsidies and carbon pricing on establishment decisions
- **Scalability:** solution quality and computation time for larger site/consumer instances

Further directions: - Stochastic programming extensions for yield and demand uncertainty - Multi-objective optimization (economic vs. ecological performance) - Integration of ecosystem service valuations (carbon sequestration, biodiversity)

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Thank You



Questions?

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